

Interview Overview final

Thursday, October 1, 2026 8:05 AM

Variable is a name used to store/reference a value in memory.

A **list** is an ordered and mutable collection that can contain duplicate elements.

Tuple is ordered and Immutable collection that can contain duplicate elements.

A **set** is a mutable collection of unique elements that does not support indexing like a list.

A **dictionary** is a mutable collection that stores data as key-value pairs.

A **function** is a reusable block of code that performs a specific task. It can accept input parameters and optionally return an output.

***args** allows a function to accept a variable number of positional arguments. Behave like tuple.

****kwargs** allows a function to accept a variable number of keyword arguments. Behave like dictionary

A **Decorator** is a function that modifies or extends the behavior of another function without changing the original function's source code.

Exception handling is a mechanism in python used to detect, handle, recover from runtime errors without abruptly terminating the program

Exception handling allows your program to handle unexpected situations gracefully instead of crashing.

An **iterator** is an object that allows you to traverse through a collection of one element at a time using the `__iter__()` and `__next__()` methods.

A **lambda function** is a small, anonymous function written in a single expression using the lambda keyword.

Map() applies a function to every element of an iterable and return iterator containing the transformed values

Filter() selects elements from an iterable based on whether a function return True or False.

Multithreading is a technique where multiple threads execute within the same process and share the same memory space.

Multiprocessing is a technique where multiple independent processes execute tasks, each with its own memory space and python interpreter.

Deep learning:

Bias: model is too simple - underfitting

Variance: model is too sensitive - overfitting

Overfitting: machine learning model learns the training data too well, including noise and unnecessary patterns.

Imagine a student who memorizes all the answers from previous exam papers but doesn't understand the concepts.

When the same questions appear:

✓ Excellent performance

When new questions appear:

✗ Poor performance

This is similar to **overfitting**.

Underfitting: When a model is too simple to learn the underlying patterns in the data.

The model performs poorly on both training and testing data.

Imagine a student who only studies a few basic concepts and doesn't understand enough of the syllabus.

When the exam comes:

✗ Poor performance on both familiar and new questions

This is similar to **underfitting**.

Activation function: introduce non-linearity allowing neural network to learn complex pattern.

Regularization is a technique that adds a penalty to the loss function to reduce overfitting and improve a model's ability to generalize.

Cross-validation is a model evaluation technique used to estimate how well a machine learning model will perform on **unseen data**.

Data Augmentation: Rotation, Scaling, Cropping, Flipping where physically valid, Brightness changes, Noise, Translation.